

COPRIME PAIRS WITH EQUAL SUM OF DIVISORS, ASSUMING SMOOTH SHIFTED PRIMES

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ABSTRACT. Let $h(x)$ be the number of pairs $1 \leq a < b < x$ with $\gcd(a, b) = 1$ and $\sigma(a) = \sigma(b)$. Erdős asked whether $h(x) > x^{2-o(1)}$ (Erdős Problem #824). By an argument of Pollack and Pomerance, this follows if for every fixed $\delta > 0$ there are $y^{1-o(1)}$ primes $p \leq 5y$ with $p+1$ free of prime factors exceeding y^δ . We write this implication out with explicit parameters and formalize it in Lean 4. We then check that the argument of the recent OpenAI preprint *Weighted dilation graphs, smooth shifted primes and totient fibers*, which proves the corresponding statement for $p-1$, applies to $p+1$ after a change of sign in two of its sections. Consequently $h(x) = x^{2-o(1)}$, conditionally on the correctness of that preprint, which has not been refereed and whose main new estimates we have not independently verified.

1. INTRODUCTION

Let $\sigma(n)$ be the sum of the divisors of n and

$$h(x) = \#\{(a, b) : 1 \leq a < b < x, \gcd(a, b) = 1, \sigma(a) = \sigma(b)\}.$$

Trivially $h(x) < x^2$. Erdős [5] asked whether $h(x)/x \rightarrow \infty$; Problem #824 of [3] asks whether $h(x) > x^{2-o(1)}$. Pollack and Pomerance [7, §6] answered Erdős's question, proving $h(x) > x^{1.4}$ for large x . Following an idea of Erdős [4], their argument shows: if $\theta > 1$ is such that there are $y^\theta/(\log y)^{O(1)}$ primes $p \leq y^\theta$ with $P^+(p+1) \leq y$, then there are at least $x^{(2-\varepsilon)(1-1/\theta)+o(1)}$ coprime pairs, for every fixed $\varepsilon > 0$. Here $P^+(m)$ is the largest prime factor of m . With the Baker–Harman value $\theta \approx 3.377$ [1] this gives the exponent 1.4; an admissible θ for every $\theta > 1$ would give $x^{2-o(1)}$, as also remarked by Tao on the problem's forum [9]. A claim of $h(x) \geq x^{13/8-o(1)}$ by a different route was submitted to the forum in August 2026 [2].

For a fixed smoothness exponent, the best unconditional result is due to Lichtman [6]: for fixed $a \neq 0$ and $\delta > 15/(32\sqrt{e}) = 0.2843\dots$ there are $\gg x/(\log x)^C$ primes $x < p \leq 2x$ with $P^+(p-a) \leq x^\delta$. The corresponding statement for every fixed $\delta > 0$ is a conjecture of Erdős. For $a = 1$ it is claimed in the OpenAI preprint [10] (“WD” below):

Theorem 1.1 ([10, Theorem 1.2]). *For every fixed $0 < \delta < 1/4$, as $x \rightarrow \infty$, $\#\{p \text{ prime} : 2x < p \leq 5x, P^+(p-1) \leq x^\delta\} \geq x^{1-o(1)}$.*

The application to σ needs $p+1$ instead of $p-1$.

Definition 1.2. Hypothesis A^+ : for every $0 < \delta < 1/4$ and every $\eta > 0$ there is y_0 such that for every real $y \geq y_0$,

$$\#\{p \text{ prime} : p \leq 5y, P^+(p+1) \leq y^\delta\} \geq y^{1-\eta}.$$

Theorem 1.3. *If A^+ holds, then for every $\varepsilon > 0$ we have $h(x) > x^{2-\varepsilon}$ for all sufficiently large x . The pairs may moreover be taken with a, b squarefree.*

Theorem 1.3 is the argument of [7, §6] with $\theta \rightarrow \infty$. We give it in full in §3, with explicit parameters and under the weaker hypothesis of $y^{1-o(1)}$ (rather than $y/(\log y)^{O(1)}$) smooth shifted primes, because this is the version we have formalized in Lean 4 (§4).

Theorem 1.4. *The proof of Theorem 1.1 in [10], modified as described in §2 and Appendix A, proves: for every fixed $0 < \delta < 1/4$, $\#\{p \text{ prime} : 2x - 2 < p \leq 5x, P^+(p+1) \leq x^\delta\} \geq x^{1-o(1)}$. In particular A^+ holds if the argument of [10] is correct.*

Corollary 1.5. *If the argument of [10] is correct, then $h(x) = x^{2-o(1)}$.*

The status of the corollary is exactly that of [10], an unrefereed preprint. Its Sections 3–5 introduce a new transference and shifted-correlation method; we have not verified them independently (§5). Our contribution is the observation that the sign change goes through, a check of the sections that it affects, and the formalization of Theorem 1.3.

2. SMOOTH SUCCESSORS FROM THE ARGUMENT OF WD

Section and theorem numbers in this section refer to [10]. WD detects primes of the form $N_u = 2u + 1$, where u carries a weight $A(u)$ supported on integers whose prime factors all lie in $[(\log x)^{20}, x^\delta]$; then $p - 1 = 2u$ is x^δ -smooth. We run the same argument with $N_u = 2u + \varsigma$ for a fixed sign $\varsigma \in \{1, -1\}$; for $\varsigma = -1$ the prime $p = 2u - 1$ has $p + 1 = 2u$. Appendix A lists every change; all other text of WD is used verbatim.

Sections 1–5 and 8 of WD are unchanged. The form $2u + 1$ does not occur in them. The shift-cancellation theorem [10, Theorem 5.1] is stated for every integral shift $0 < |k| \leq L^C$ ($L = \log x$), and the physical operator underlying the transference theorem [10, Theorem 3.5] moves $n \mapsto n + kD$ with $0 < |k|$. We checked that WD uses no distribution result for primes in the residue class ± 1 modulo d (of Bombieri–Vinogradov type or otherwise): primes enter only through the Siegel–Walfisz theorem in all reduced residue classes and the prime number theorem in short logarithmic cells.

Section 6 (Type II, [10, Theorem 6.1]). The estimate concerns $\sum_{mn=2u+\varsigma} A(u)\Psi(u/x)\alpha_m\beta_n$. After the split $u = eh$ and Cauchy’s inequality in (e, n) , two factorizations with common e, n ,

$$mn = 2eh + \varsigma, \quad m'n = 2eh' + \varsigma, \quad m \neq m',$$

force $2e \mid m' - m$, because $mn \equiv \varsigma \pmod{2e}$ makes n a unit modulo $2e$. Write $m' - m = 2\varsigma ek$ with $k \neq 0$. Then $h' - h = \varsigma kn$ and

$$mh' - m'h = m(h + \varsigma kn) - (m + 2\varsigma ek)h = \varsigma k(mn - 2eh) = \varsigma^2 k = k,$$

so $z = m'h$ and $z + k = mh'$ are exactly the endpoints of WD. Conversely, given $k \neq 0$, $m'h = z$, $mh' = z + k$, $m' \equiv m \pmod{2|k|}$ and $e = (m' - m)/(2\varsigma k) > 0$, we get $m(h' - h) = \varsigma k(2eh + \varsigma)$. As every prime factor of m exceeds W while $|k| \leq L^{O(1)}$, $(m, k) = 1$; so $n = (2eh + \varsigma)/m$ is a positive integer ($2eh + \varsigma \geq 1$) and both factorizations are recovered. The diagonal becomes $\sum_v \tau(v)\tau(v - \varsigma)^3$, bounded exactly as before, and the pulled-back cutoff $n = (m' - m)z/(\varsigma kmm')$ or ς/m has the same derivative bounds. Everything after the endpoint identification is unchanged.

WD’s own Type II argument already uses negative shifts: there $m' - m = 2ek$ with $e > 0$, so $k < 0$ whenever $m' < m$, and both orders occur in the off-diagonal sum. Hence the $p + 1$ version needs [10, Theorem 5.1] for no shifts other than those WD itself uses.

Section 7 (prime extraction). Put $N_u = 2u + \varsigma$.

- Type I: for odd squarefree d , $2u + \varsigma \equiv 0 \pmod{d}$ means $u \equiv -\varsigma 2^{-1} \pmod{d}$, a single reduced class. The proof uses only that the class is a fixed reduced class.
- Preliminary sieve: odd primes divide N_u with density $1/(p-1)$ for either sign.
- Balanced composites: $D_1 \mid u$ gives $mn \equiv \varsigma \pmod{D_1}$, and the banned event at a sieve prime l is $mn \equiv \varsigma \pmod{l}$. It has $l-1$ residue pairs and is disjoint from $mn \equiv 0 \pmod{l}$ ($2l-1$ pairs); there are $\varphi(D_1)$ pairs modulo D_1 with $mn \equiv \varsigma$. All densities and remainders are as in WD.
- Conclusion: since Ψ is supported in $[1, 2]$, the primes $p = 2u + \varsigma$ satisfy $2x - 1 \leq p \leq 4x + 1$, and $P^+(p - \varsigma) = P^+(2u) \leq x^\delta$.

3. PROOF OF THEOREM 1.3

Write $L = \log x$. Fix $\varepsilon \in (0, 1)$, a real $B \geq 8/\varepsilon$, $\eta = \varepsilon/16$ and $\delta = 1/(2B) \in (0, 1/4)$; these are fixed before $x \rightarrow \infty$. For large x put $y = L^B$,

$$\mathcal{P} = \{p \text{ prime} : p \leq 5y, P^+(p+1) \leq y^\delta\}, \quad M = |\mathcal{P}|.$$

By A^+ , $M \geq y^{1-\eta} = L^{B(1-\eta)}$ for all large x . Let $k = \lfloor L/\log(5y+1) \rfloor$. Then $(5y+1)^k \leq x$, and since $\log(5y+1) = B \log L + O(1)$,

$$(1) \quad k = \frac{L}{B \log L} (1 + o(1)), \quad k \leq L.$$

For large x , $M \geq L^2 \geq 2k$ because $B(1-\eta) > 2$.

3.1. The family. Let $\mathcal{A}$ be the set of products of k distinct primes of $\mathcal{P}$. Each $a \in \mathcal{A}$ is squarefree and $a < (5y+1)^k \leq x$, and distinct k -subsets give distinct products. So $N := |\mathcal{A}| = \binom{M}{k} \geq (M/k)^k$ and, by (1) and $\log k \leq \log L$,

$$(2) \quad \log N \geq k \log L (B(1-\eta) - 1) = L \left(1 - \eta - \frac{1}{B}\right) (1 + o(1)).$$

For $a = p_1 \cdots p_k$ we have $\sigma(a) = \prod_i (p_i + 1) \leq x$, and every prime factor of $\sigma(a)$ is at most $y^\delta = L^{1/2}$. The number U of $L^{1/2}$ -smooth integers in $[1, x]$ satisfies

$$(3) \quad U \leq \left(1 + \frac{L}{\log 2}\right)^{L^{1/2}}, \quad \log U = o(L),$$

since such an integer is $\prod_{q \leq L^{1/2}} q^{e_q}$ with $0 \leq e_q \leq L/\log 2$.

3.2. Many equal values. With $r(s) = \#\{a \in \mathcal{A} : \sigma(a) = s\}$, Cauchy's inequality gives

$$\#\{(a, b) \in \mathcal{A}^2 : \sigma(a) = \sigma(b)\} = \sum_s r(s)^2 \geq N^2/U.$$

3.3. Pairs sharing many primes. Let $t = \lceil \eta k \rceil$. The ordered pairs in $\mathcal{A}^2$ sharing at least t primes number at most $N \binom{k}{t} \binom{M}{k-t}$ (choose a , t of its primes, and the other $k-t$ primes of b), and

$$\binom{M}{k-t} / \binom{M}{k} = \prod_{i=0}^{t-1} \frac{k-i}{M-k+t-i} \leq \left(\frac{2k}{M}\right)^t, \quad \binom{k}{t} \leq 2^k.$$

By (1), (3) and $M \geq L^{B(1-\eta)}$,

$$\begin{aligned} \log(2^k (2k/M)^t \cdot 2U) &\leq (k+1) \log 2 + \log U - \eta k (B(1-\eta) \log L - \log(2L)) \\ &= -\eta L \left(1 - \eta - \frac{1}{B}\right) (1 + o(1)) + o(L), \end{aligned}$$

which is negative for large x . So at least $N^2/(2U)$ ordered pairs (a, b) have $\sigma(a) = \sigma(b)$ and share fewer than t primes; in particular $a \neq b$.

3.4. Removing the common factor. For such a pair let $g = \gcd(a, b)$, $u = a/g$, $v = b/g$. As a, b are squarefree, g, u, v are pairwise coprime, so $\sigma(g)\sigma(u) = \sigma(g)\sigma(v)$ and $\sigma(u) = \sigma(v)$, with $\gcd(u, v) = 1$, $u \neq v$, and $u, v < x$ squarefree. The integer g is a product of at most $t - 1 < \eta k$ primes below $5y + 1$, so $g \leq (5y + 1)^{\eta k} \leq x^\eta$. Given the ordered pair (u, v) , the pair $(a, b) = (gu, gv)$ is determined by g ; each unordered pair $\{u, v\}$ therefore arises from at most $2x^\eta$ ordered pairs (a, b) . Hence

$$h(x) \geq \frac{N^2}{4Ux^\eta}, \quad \log h(x) \geq \left(2 - 3\eta - \frac{2}{B}\right)L - o(L) > (2 - \varepsilon)L$$

for large x , because $3\eta + 2/B \leq 3\varepsilon/16 + \varepsilon/4 < \varepsilon/2$. $\square$

Remark 3.1. Since $h(x)$ only depends on the integers below x , $h(x) = h(\lceil x \rceil)$; so it suffices to prove the theorem for integer x , which is the form formalized in Lean. Pollack and Pomerance count pairs with $b \leq x$; the difference is immaterial.

4. FORMALIZATION

Theorem 1.3 is formalized in Lean 4 (v4.33.1) with Mathlib (commit [0df444a](#)) in the directory `lean/` of [8]. The definitions are

```
noncomputable def h (x : ℕ) : ℕ :=
  ((Finset.Ico 1 x ×ˢ Finset.Ico 1 x).filter
    (fun p => p.1 < p.2 ∧ Nat.Coprime p.1 p.2 ∧ sigma 1 p.1 = sigma 1 p.2)).card
def SmoothSuccessorHypWeak : Prop :=
  ∀ δ : ℝ, 0 < δ → δ < 1/4 → ∀ η : ℝ, 0 < η → ∃ y₀ : ℝ, ∀ y : ℝ, y₀ ≤ y →
    (y ^ (1 - η) : ℝ) ≤ (((Finset.Iic [5*y]₊).filter
      (fun p => p.Prime ∧ ∀ q ∈ (p+1).primeFactors, ((q:ℕ):ℝ) ≤ y ^ δ)).card : ℝ)
```

where `sigma 1` is Mathlib's $\sigma_1 = \sigma$; `SmoothSuccessorHypWeak` is Definition 1.2. The theorem

```
theorem erdos824_of_smoothSuccessorWeak (H : SmoothSuccessorHypWeak) :
  ∀ ε : ℝ, 0 < ε → ∃ X : ℝ, ∀ x : ℕ, X ≤ x → (x:ℝ) ^ (2 - ε) < (h x : ℝ)
```

has no `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The Lean proof follows §3 with $B = 16/\varepsilon$ and $\delta = \varepsilon/32$.

5. STATUS

Theorem 1.3 is unconditional and machine-checked. Theorem 1.4 rests on [10]. We rechecked Sections 6–7 of WD for the change of sign, but we have not independently verified Sections 3–5 (transference for weighted dilation graphs, construction of the ideal operator, lifted shift cancellation; about 2800 lines of source). Writing Lean statements for WD exposed several underspecified points. Examples: the hypothesis $\ell \geq 1$ is used in Section 4 but not stated in the definition of prime groups; Section 5 uses a coefficient form of the ideal-operator theorem [10, Theorem 4.1] that the theorem does not state. None is known to affect the conclusion.

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The reduction in §3 is due to Pollack and Pomerance [7], after Erdős; its relevance to Problem #824 was also pointed out by T. Tao. AI tools were used extensively in this work. Claude (Anthropic) adapted [10] to the sign ς , wrote the text and the Lean formalization. GPT models (OpenAI) were used to referee drafts. The preprint [10] is itself AI-generated. The author has checked the arguments presented here and takes responsibility for them.

APPENDIX A. LIST OF CHANGES TO WD

Labels are those of the L^AT_EX source of [10]; all other text of WD is used verbatim.

Location in WD	original	with sign ς
Thm 6.1, <code>eq:typeii-bound</code>	$mn = 2u + 1$	$mn = 2u + \varsigma$
§6, pre-Cauchy bound, divisor count	$\tau(2u + 1)$, divisors of $2u + 1$	$\tau(2u + \varsigma)$, divisors of $2u + \varsigma$
§6, <code>eq:typeii-separated-B</code> , <code>eq:typeii-cauchy</code>	$mn = 2eh + 1$	$mn = 2eh + \varsigma$
§6, <code>eq:typeii-diagonal</code>	$e \mid (mn - 1)/2, \tau(v - 1)^3$	$e \mid (mn - \varsigma)/2, \tau(v - \varsigma)^3$
§6, <code>eq:typeii-determinant</code>	$m' - m = 2ek, h' - h = kn$	$m' - m = 2\varsigma ek, h' - h = \varsigma kn$
§6, <code>eq:typeii-converse-identity</code>	$m(h' - h) = k(2eh + 1)$	$m(h' - h) = \varsigma k(2eh + \varsigma)$
§6, <code>eq:typeii-pulled-back</code>	$e = \frac{m' - m}{2k}, n = \dots + \frac{1}{m}$	$e = \frac{m' - m}{2\varsigma k}, n = \dots + \frac{\varsigma}{m}$
§7, definition	$N_u = 2u + 1$	$N_u = 2u + \varsigma$
§7, <code>prop:typeI</code>	class $2u + 1 \equiv 0$	class $u \equiv -\varsigma 2^{-1} \pmod{d}$
§7, balanced range	$u = \frac{mn-1}{2}, mn \equiv 1 \pmod{D_1}$, banned $mn \equiv 1$	$u = \frac{mn-\varsigma}{2}, mn \equiv \varsigma \pmod{D_1}$, banned $mn \equiv \varsigma$
§7, conclusion	$2u + 1$ prime, $P^+(p - 1)$	$2u + \varsigma$ prime, $P^+(p - \varsigma)$

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